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Nakamura et al.

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(54) **SHEET CONVEYANCE DEVICE AND IMAGE FORMING APPARATUS**

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Primary Examiner — Jeremy R Severson

(74) *Attorney, Agent, or Firm* — Merchant & Gould P.C.

(71) Applicant: **BROTHER KOGYO KABUSHIKI**
KAISHA, Nagoya (JP)

(72) Inventors: **Kazutoshi Nakamura**, Kuwana (JP);
Tomonori Watanabe, Ichinomiya (JP)

(73) Assignee: **BROTHER KOGYO KABUSHIKI**
KAISHA, Nagoya (JP)

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G03G 15/00 (2006.01)

(52) **U.S. Cl.**
CPC **G03G 15/6529** (2013.01); **B65H 2403/48** (2013.01); **B65H 2403/50** (2013.01); **B65H 2403/72** (2013.01); **B65H 2601/11** (2013.01); **B65H 2601/325** (2013.01); **B65H 2801/27** (2013.01); **G03G 2221/1657** (2013.01)

(58) **Field of Classification Search**
CPC B65H 3/0669; B65H 2403/72; B65H 2403/942; G03G 15/20–15/2098
See application file for complete search history.

(57) **ABSTRACT**

A transmission gear is movable in an axial direction between a first position separated from an output gear and a second position coupled to the output gear in a state where the transmission gear engages with an input gear. The input gear is configured to: by rotating in a first direction by receiving driving force from a drive source, move the transmission gear from the first position to the second position and transmit driving force in the first direction to the output gear via the transmission gear; and by rotating in a second direction, move the transmission gear from the second position to the first position and cancel transmission of driving force to the output gear. The transmission gear is configured to, when the output gear rotates in a state where the transmission gear is located at the first position, not be coupled to the output gear.

18 Claims, 12 Drawing Sheets

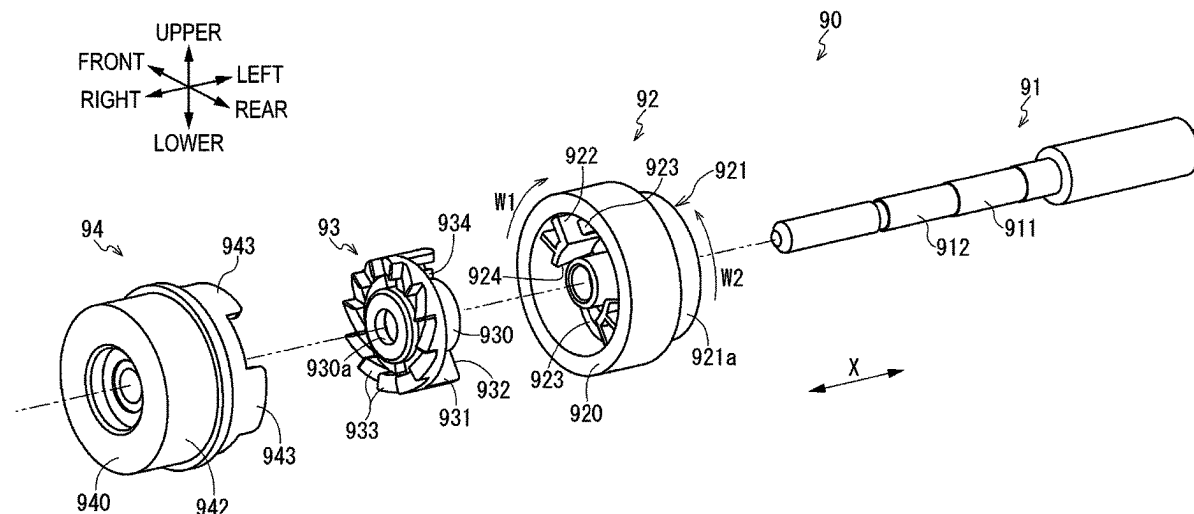


FIG. 2

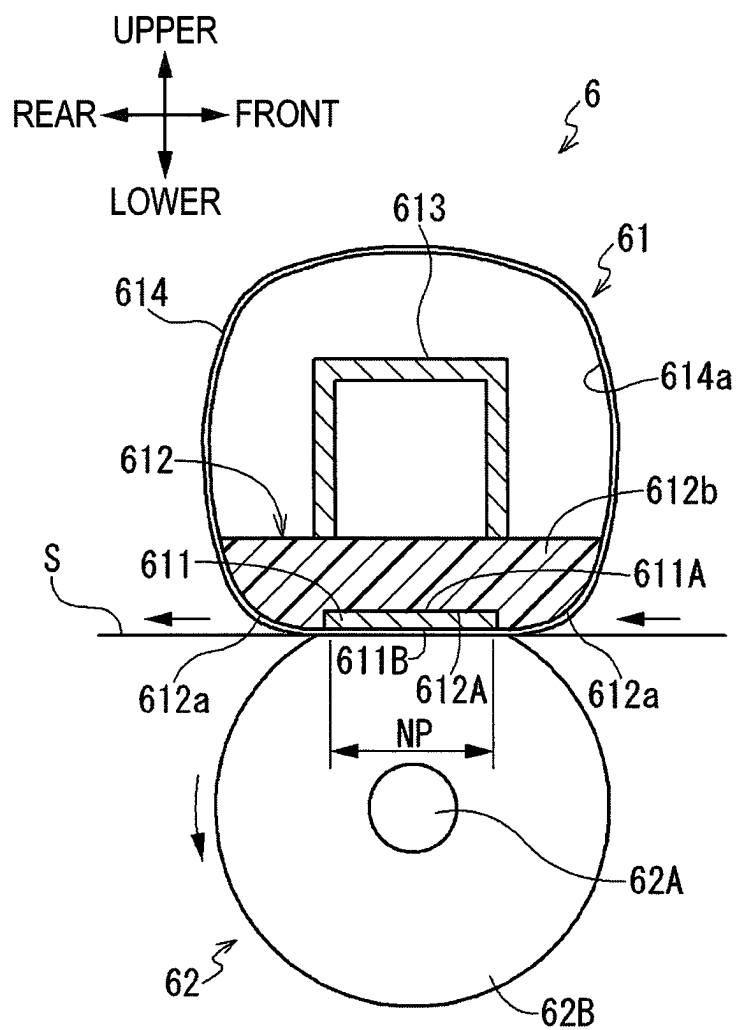


FIG. 3

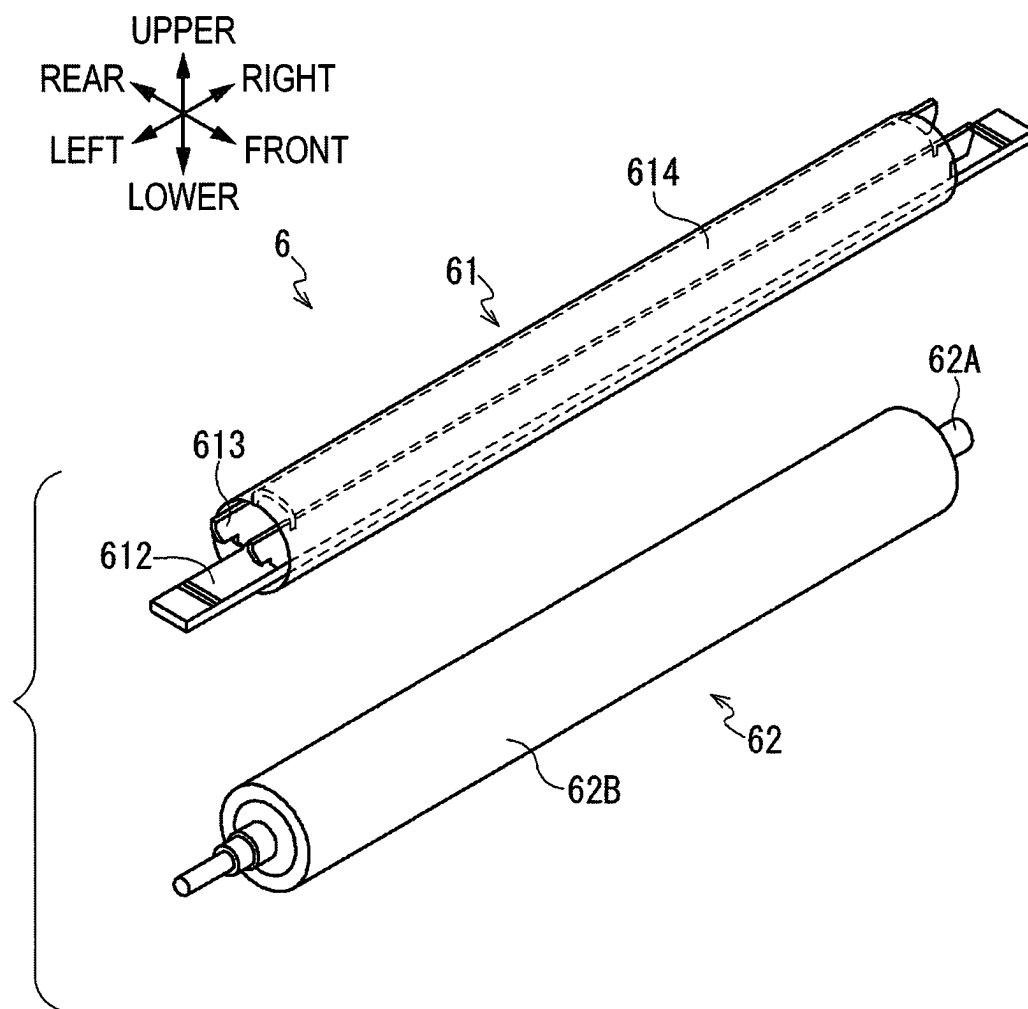


FIG. 4

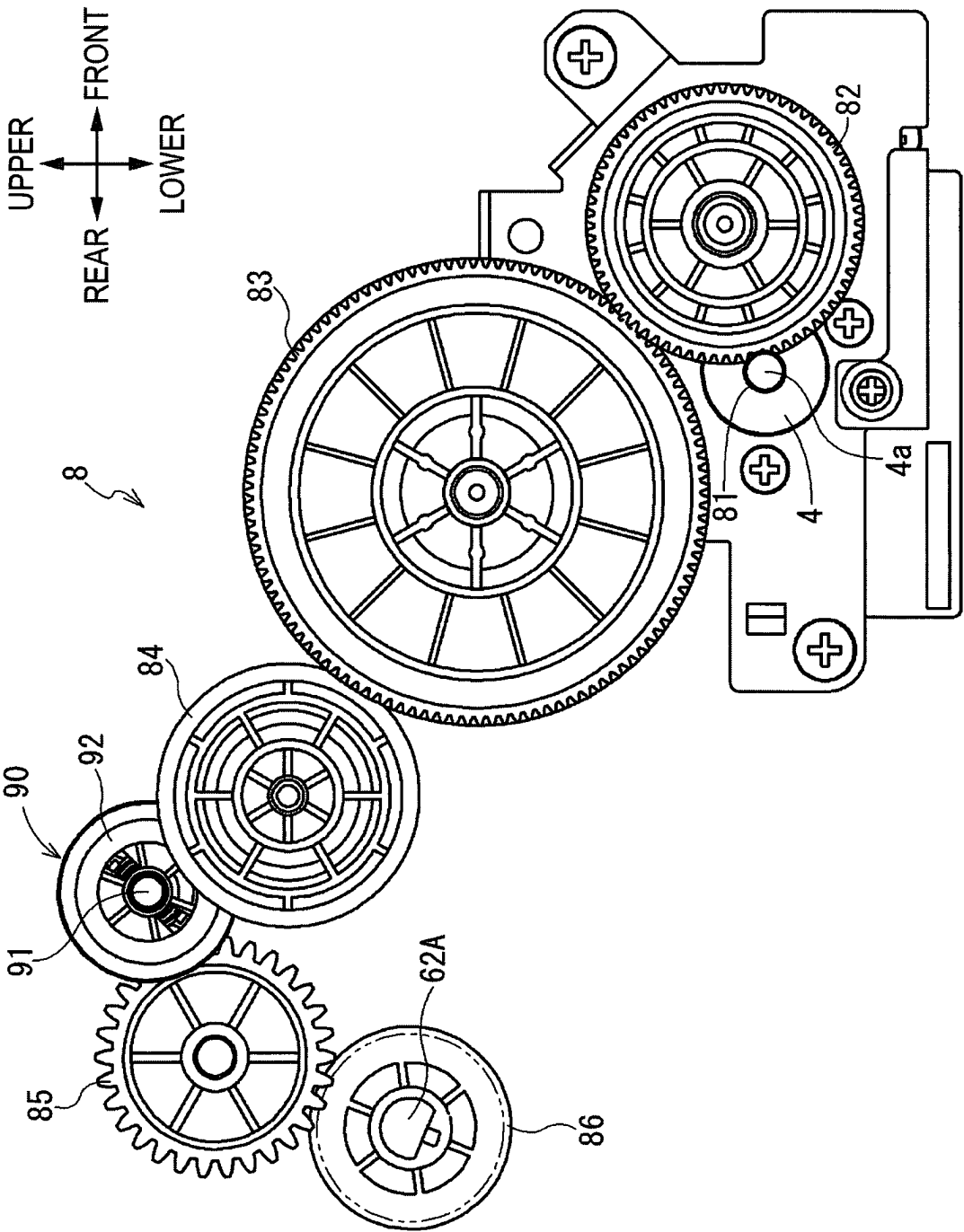
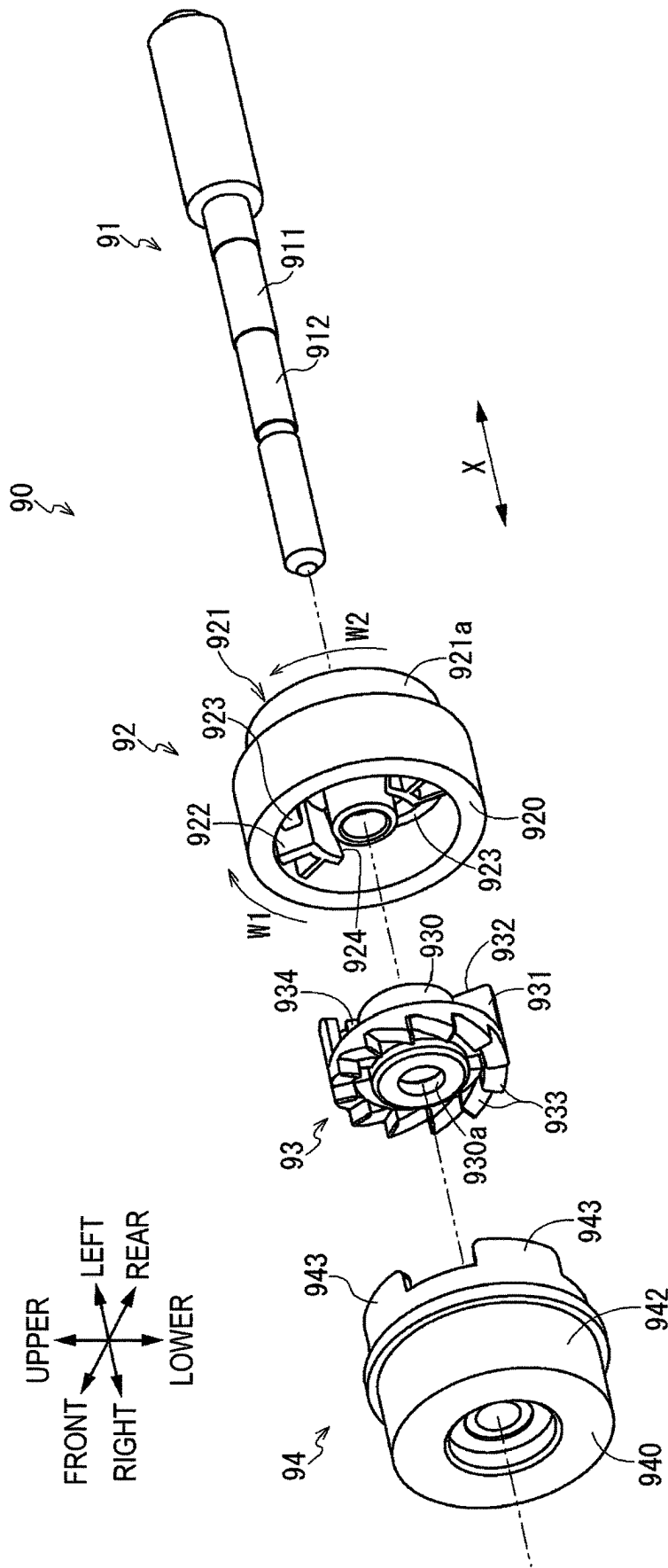


FIG. 5



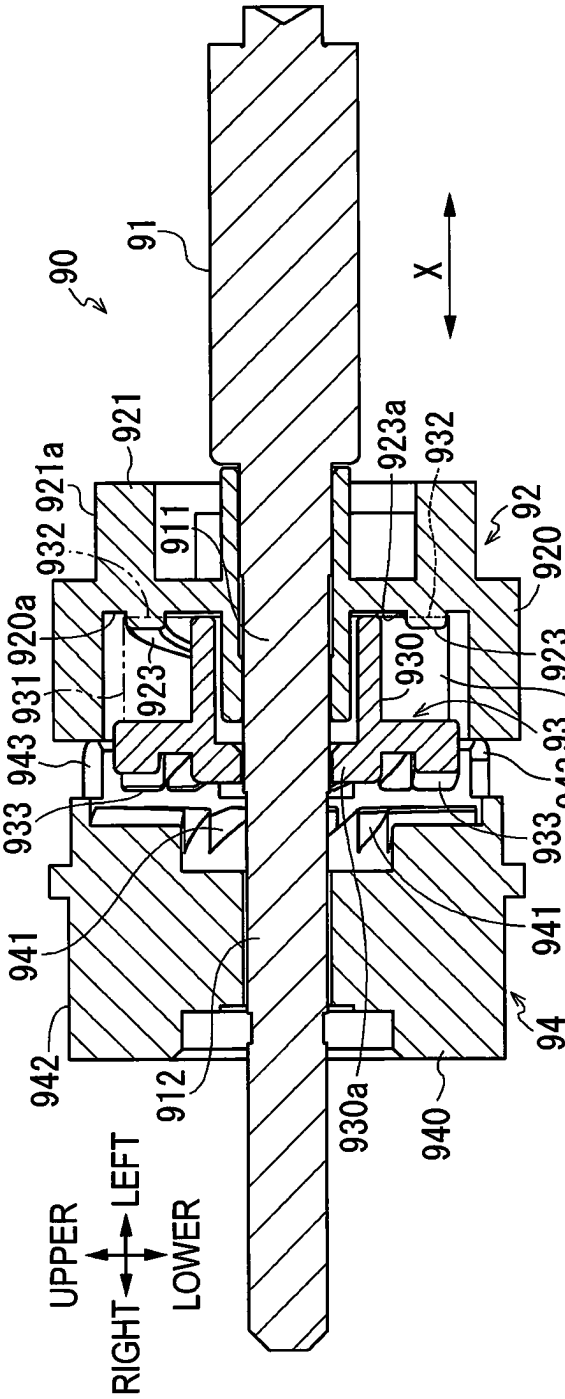


FIG. 6A

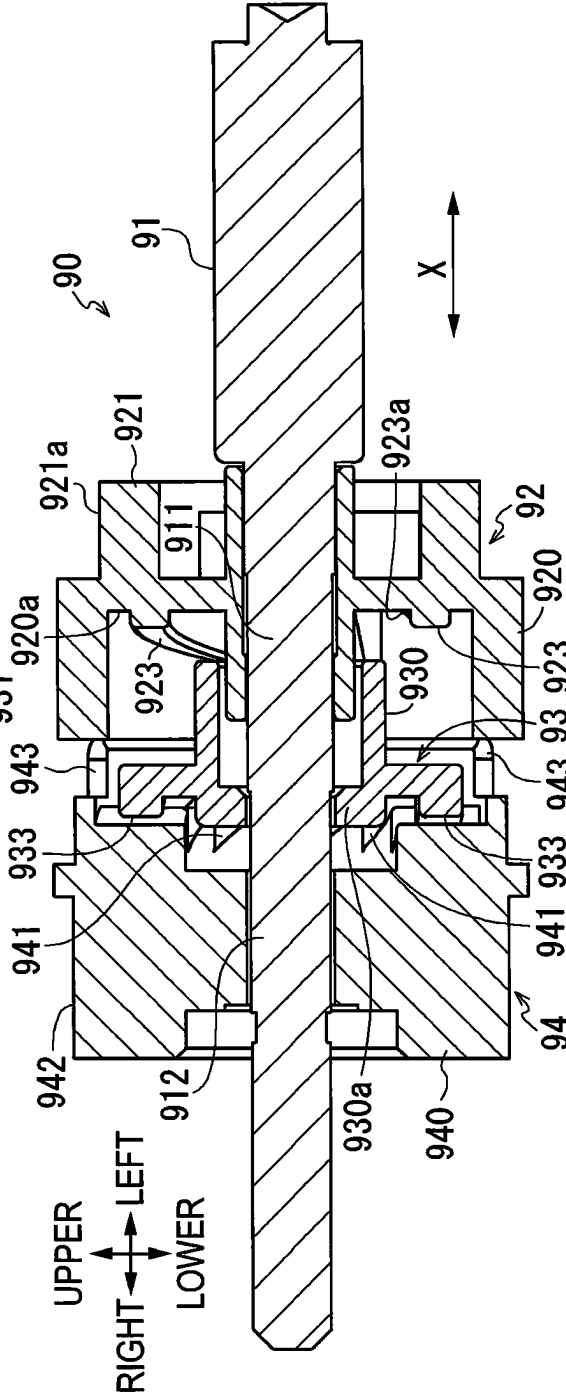


FIG. 6B

FIG. 7

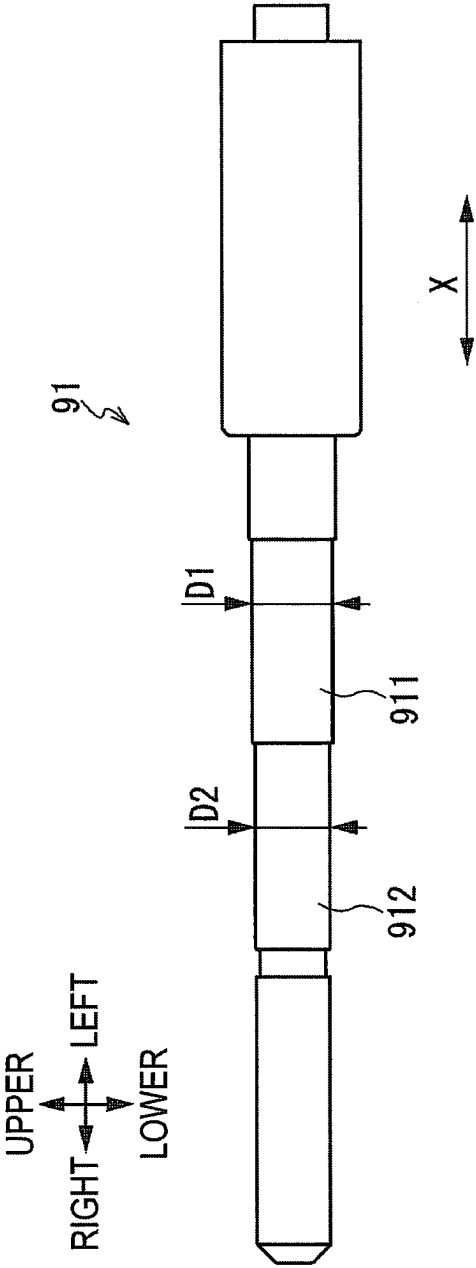


FIG. 8A

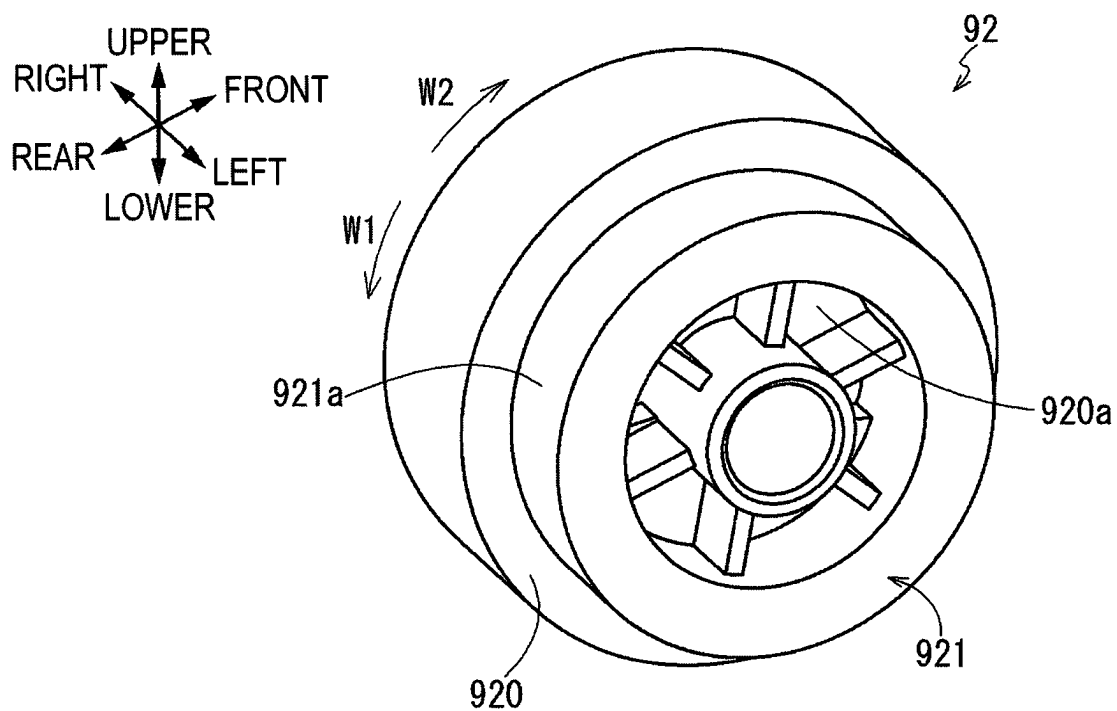


FIG. 8B

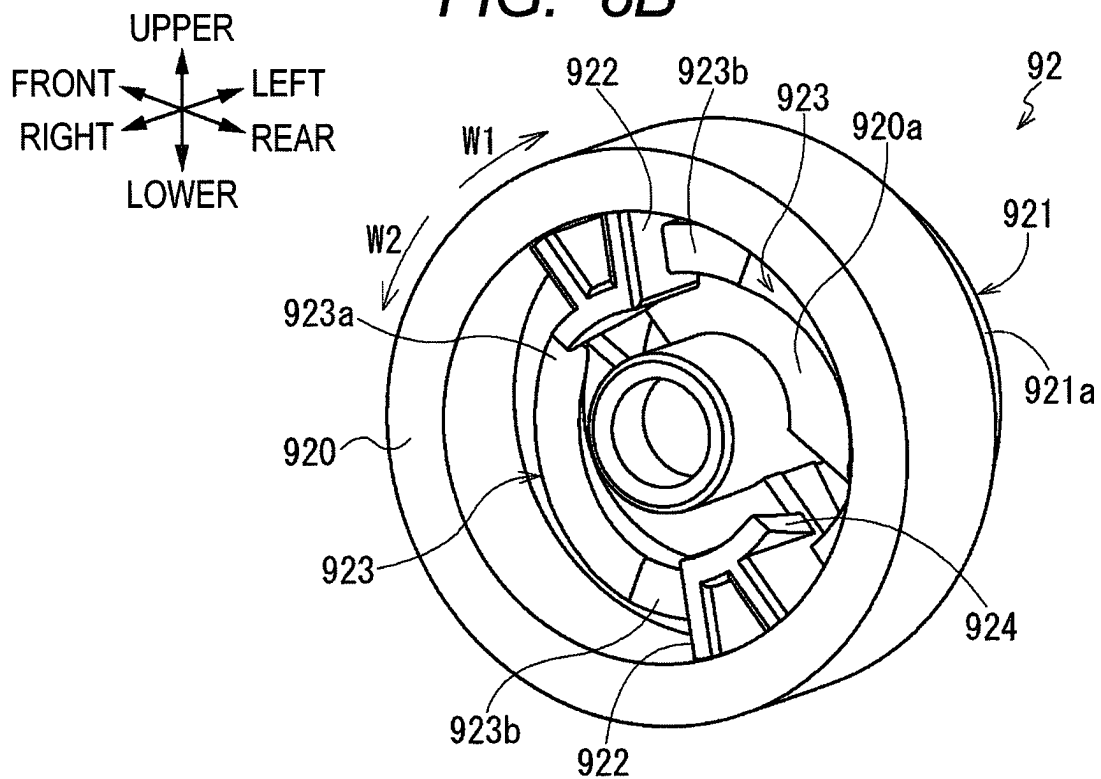


FIG. 9

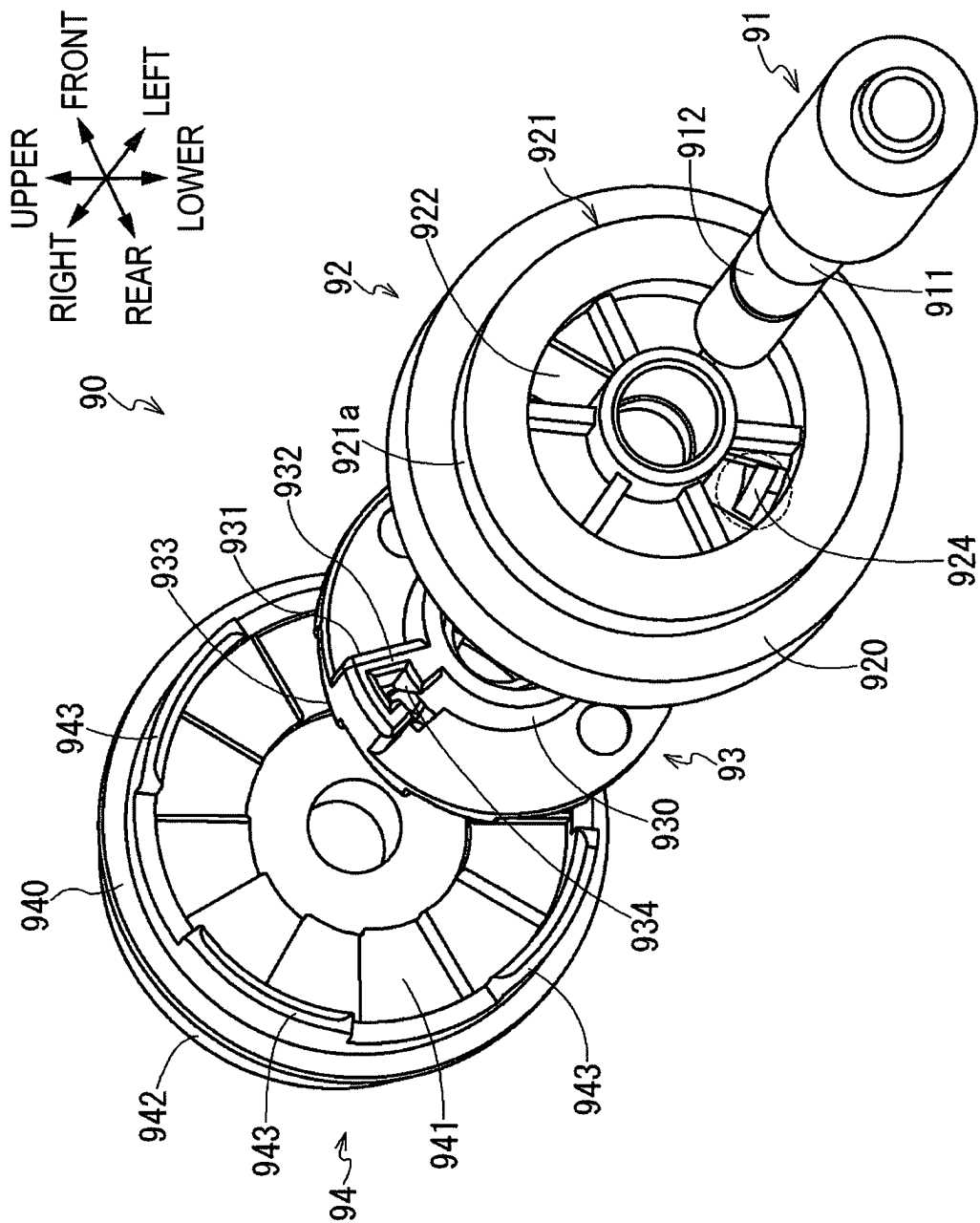


FIG. 10A

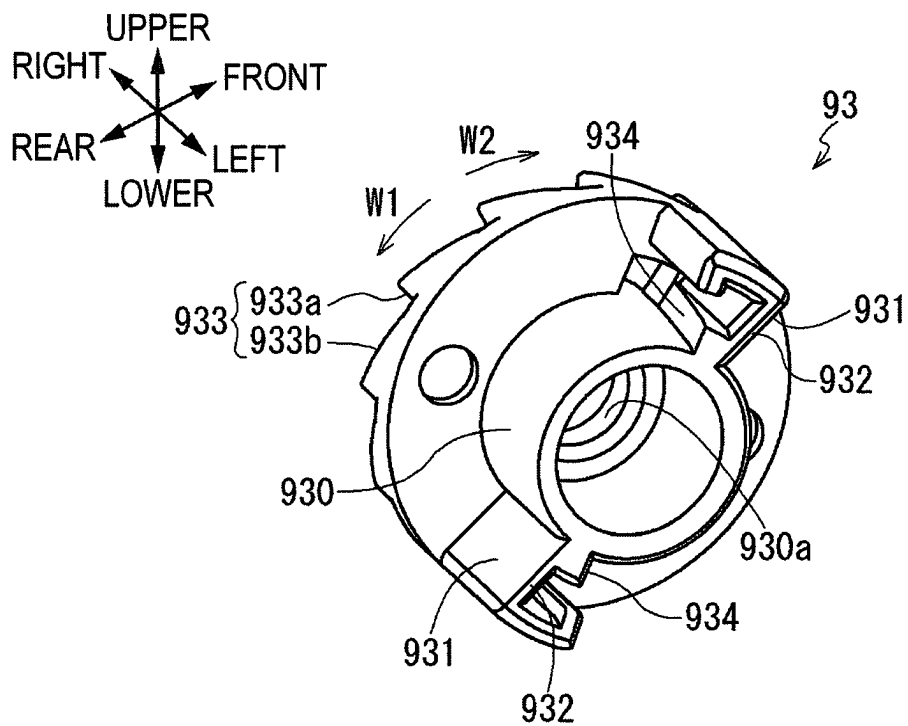


FIG. 10B

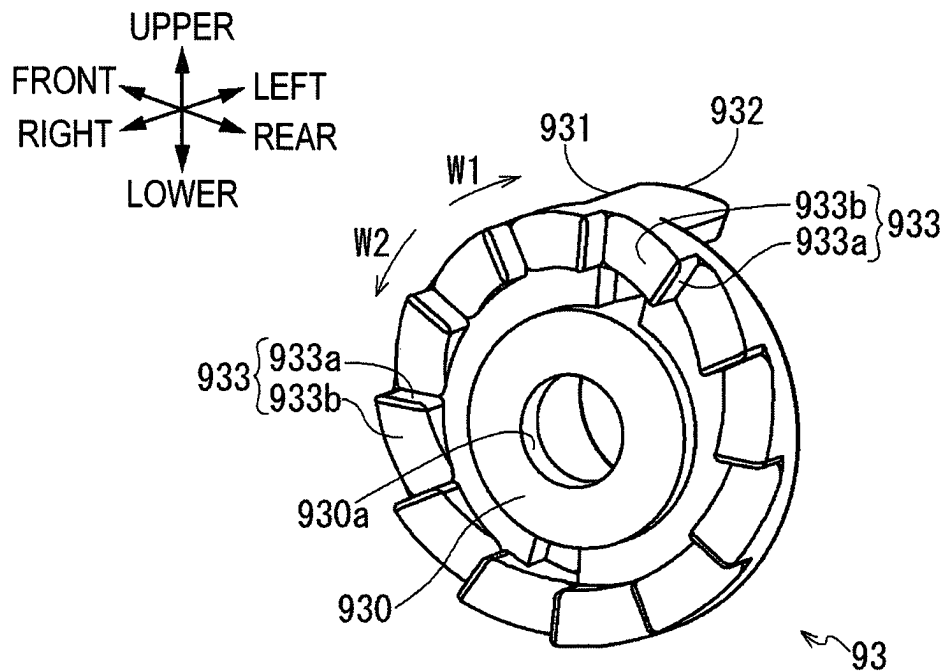


FIG. 11A

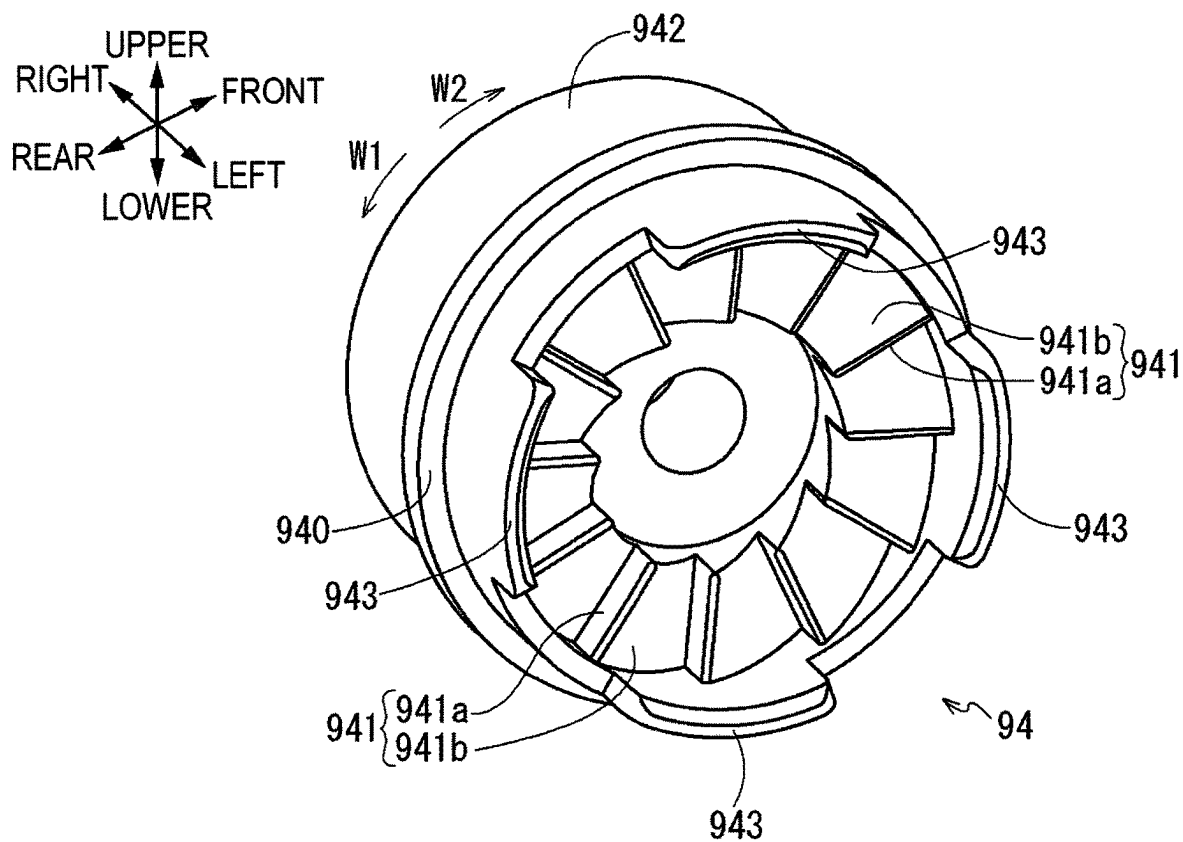


FIG. 11B

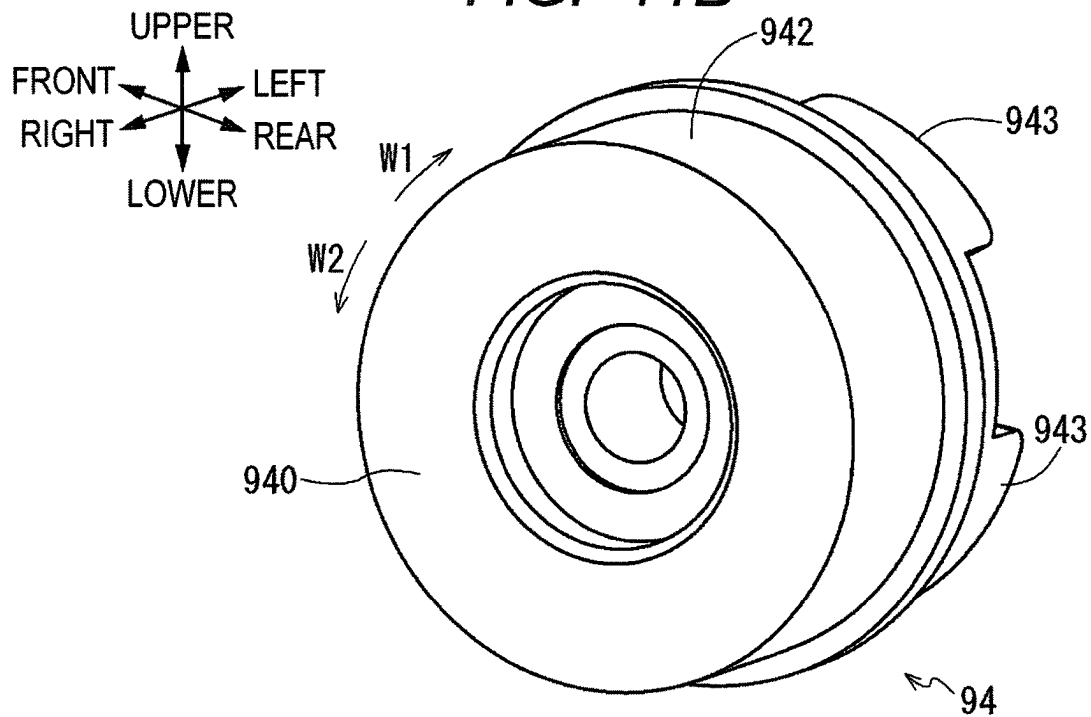
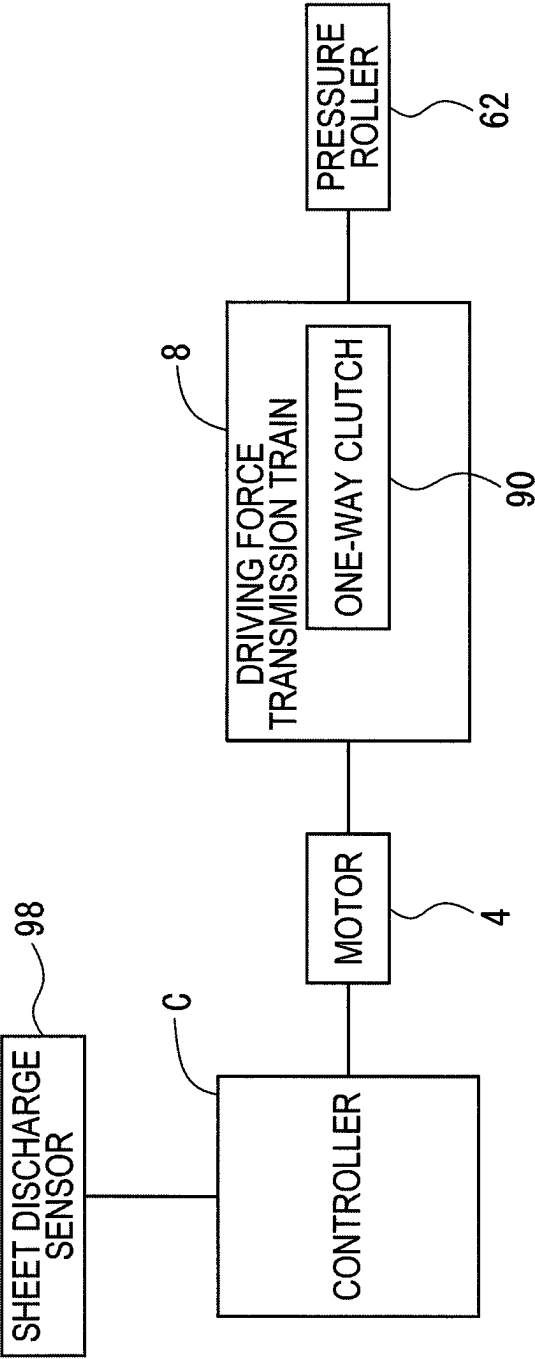


FIG. 12



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SHEET CONVEYANCE DEVICE AND IMAGE FORMING APPARATUS

REFERENCE TO RELATED APPLICATIONS

This application claims priority from Japanese Patent Application No. 2021-192563 filed on Nov. 26, 2021. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

A one-way clutch that transmits rotational force in only one direction is conventionally known. The one-way clutch is used, for example, in a sheet conveyance device provided in an image forming apparatus.

DESCRIPTION

In a sheet conveyance device, a one-way clutch is incorporated in a drive train that transmits drive power from a motor to a fuser (fixing device) that conveys a sheet and fixes an image on the sheet.

In such a fuser, a sheet jam may occur between fixing rollers. When a jam occurs, it is necessary to perform a jam removal process of pulling out the jammed sheet by hand. At this time, if the fixing roller and the motor are connected to each other during the jam removal process, the load of the motor and the drive train is applied to rotation of the fixing roller, which may hinder a smooth jam removal process.

In order to deal with the above problem, it is conceivable to provide a driving force transmission release mechanism in addition to a one-way clutch, in driving force transmission means for transmitting driving force from a drive source to the fixing roller. The driving force transmission release mechanism performs transmission and release of driving force between the fixing roller and the drive source, thereby releasing coupling between the fixing roller and the drive source during a jam removal process. However, if such a driving force transmission release means is provided, the configuration of the driving force transmission means may become complicated.

In view of the foregoing, an example of an object of this disclosure is to provide a sheet conveyance device and an image forming apparatus configured to enable a smooth jam removal process without providing a complicated mechanism in a driving force transmission means.

According to one aspect, this specification discloses a sheet conveyance device. The sheet conveyance device includes a drive source, a roller configured to convey a sheet, and a one-way clutch configured to transmit a driving force from the drive source to the roller. The one-way clutch includes a fixed shaft, an input gear, an output gear, and a transmission gear. The fixed shaft extends in an axial direction. The input gear is rotatably supported by the fixed shaft. The input gear is rotatable by receiving the driving force from the drive source. The output gear is rotatably supported by the fixed shaft. The output gear is coupled to the roller and configured to transmit the driving force to the roller. The transmission gear is supported by the fixed shaft. The transmission gear is movable in the axial direction between a first position and a second position in a state where the transmission gear engages with the input gear. The first position is a position at which the transmission gear is separated from the output gear. The second position is a position at which the transmission gear is coupled to the output gear. The input gear is configured to: by rotating in a

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first direction by receiving the driving force from the drive source, move the transmission gear from the first position to the second position and transmit the driving force in the first direction to the output gear via the transmission gear; and by rotating in a second direction opposite the first direction by receiving the driving force from the drive source, move the transmission gear from the second position to the first position and cancel transmission of the driving force to the output gear. The transmission gear is configured to, when the output gear rotates in a state where the transmission gear is located at the first position, not be coupled to the output gear. According to another aspect, this specification also discloses an image forming apparatus including a print engine and the above-described sheet conveyance device.

According to the above-described configuration, when the output gear rotates in a state where the transmission gear is located at the first position, the transmission gear is not coupled to the output gear. Because the transmission gear and the output gear are not coupled, the roller is easily rotated when performing a jam removal process of removing a jammed sheet without providing a complicated mechanism in driving force transmission means for transmitting the driving force from the drive source to the roller. Thus, the jam removal process is performed smoothly.

FIG. 1 is a central cross-sectional view showing an image forming apparatus.

FIG. 2 is a side cross-sectional view showing a fuser.

FIG. 3 is a perspective view showing a heating unit and a pressure roller of the fuser;

FIG. 4 is a side view showing a driving force transmission train.

FIG. 5 is an exploded perspective view showing a one-way clutch.

FIG. 6A is a side cross-sectional view showing the one-way clutch in a state where a transmission gear is located at a first position.

FIG. 6B is a side cross-sectional view showing the one-way clutch in a state where the transmission gear is located at a second position.

FIG. 7 is a rear view showing a fixed shaft of the one-way clutch.

FIGS. 8A and 8B are perspective views showing an input gear of the one-way clutch.

FIG. 9 is an exploded perspective view of the one-way clutch in which an inclined engagement surface of the input gear is visibly shown.

FIGS. 10A and 10B are perspective views showing a transmission gear of the one-way clutch.

FIGS. 11A and 11B are perspective views showing an output gear of the one-way clutch.

FIG. 12 is a block diagram showing a controller to which a motor and a sheet discharge sensor are connected.

Next, a mode for carrying out the present disclosure will be described with reference to the accompanying drawings. [Image Forming Apparatus]

An image forming apparatus 1 according to an embodiment shown in FIG. 1 is provided with a sheet conveyance device. For example, the image forming apparatus 1 is a laser printer that forms an image on a sheet S by an electrophotographic method.

In the following description, the right side in FIG. 1 is defined as the front side of the image forming apparatus 1, the left side in FIG. 1 is defined as the rear side of the image forming apparatus 1, the near side in the direction perpendicular to the drawing sheet of FIG. 1 is defined as the left side of the image forming apparatus 1, and the far side in the direction perpendicular to the drawing sheet of FIG. 1 is

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defined as the right side of the image forming apparatus 1. Further, the upper side and the lower side in FIG. 1 are defined as the upper side and the lower side of the image forming apparatus 1, respectively.

The image forming apparatus 1 includes an apparatus main body 2, a sheet feeder 3, a print engine (image forming section) 5, a fuser (fixing device) 6, a sheet discharge section 7, and a sheet conveyance device T.

The apparatus main body 2 accommodates the sheet feeder 3, the print engine 5, the fuser 6, the sheet discharge section 7, and the sheet conveyance device T.

The sheet feeder 3 includes a sheet feed tray 10 that supports the sheet S, a sheet conveyor 30, a conveyance roller pair 34, and a registration roller pair 35. The sheet feeder 3 is arranged in the lower part of the apparatus main body 2 and conveys the sheet S supported by the sheet feed tray 10 to the print engine 5. The image forming apparatus 1 has a conveyance path P for the sheet S from the sheet feeder 3 to the sheet discharge section 7 via the print engine 5.

The sheet feed tray 10 has a pressure plate 12 and a pressing plate 13. The pressure plate 12 is a plate member that supports the sheet S from below. The pressure plate 12 is rotatable about a pivot point 12a at the rear end thereof, and is movable up and down between a lowered position and a raised position by pivoting about the pivot point 12a. The pressing plate 13 is located below the pressure plate 12, and causes the pressure plate 12 to move up and down between the lowered position and the raised position.

The sheet conveyor 30 is a conveyance mechanism that separates and picks up the sheets S supported by the sheet feed tray 10 one sheet at a time and conveys the sheet S toward the print engine 5. The sheet conveyor 30 includes a sheet feed roller 31, a separation roller 32, and a separation pad 33.

The sheet feed roller 31 is a roller for sending the sheet S supported by the sheet feed tray 10 toward the separation roller 32. The separation roller 32 is arranged downstream of the sheet feed roller 31 in a sheet conveyance direction. The separation pad 33 is arranged to face the separation roller 32 and is urged toward the separation roller 32.

The sheets S sent out toward the separation roller 32 by the sheet feed roller 31 are separated to one sheet between the separation roller 32 and the separation pad 33. The sheet S separated to one sheet is sent to a conveyance path P.

The sheet S sent to the conveyance path P is conveyed toward the print engine 5 by the conveyance roller pair 34 and the registration roller pair 35. The registration roller pair 35 regulates the movement of the leading edge of the conveyed sheet S to temporarily stop the sheet S, and then conveys the sheet S toward the print engine 5 at a particular timing.

The print engine 5 is arranged downstream of the sheet feeder 3 in the sheet conveyance direction, and forms an image on the sheet S conveyed from the sheet feeder 3. The print engine 5 includes a process cartridge 50 that transfers an image onto the surface of the sheet S conveyed from the sheet feeder 3 and an exposure unit 56 that exposes the surface of a photosensitive drum 54 in the process cartridge 50.

The process cartridge 50 is arranged above the sheet feeder 3 in the apparatus main body 2, and includes a developer storage chamber 51, a supply roller 52, a development roller 53, the photosensitive drum 54, a transfer roller 55, and so on.

The exposure unit 56 includes a laser diode, a polygon mirror, a lens, a reflecting mirror, and so on. The exposure

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unit 56 irradiates the photosensitive drum 54 with laser light based on image data inputted to the image forming apparatus 1, thereby exposing the surface of the photosensitive drum 54.

The developer storage chamber 51 stores toner serving as a developer. The toner stored in the developer storage chamber 51 is sent to the supply roller 52 while being stirred by an agitator (not shown). The supply roller 52 further supplies the toner sent from the developer storage chamber 51 to the development roller 53.

The development roller 53 is arranged in close contact with the supply roller 52, and bears toner supplied from the supply roller 52 and positively charged by a sliding contact member (not shown). A developing bias is applied to the development roller 53 by bias applying means (not shown).

The photosensitive drum 54 is arranged adjacent to the development roller 53. The surface of the photosensitive drum 54 is uniformly charged by a charger (not shown) and then exposed by the exposure unit 56. The exposed portion of the photosensitive drum 54 has a lower potential than the other portions, and an electrostatic latent image is formed on the photosensitive drum 54 based on the image data. Then, positively charged toner is supplied from the development roller 53 to the surface of the photosensitive drum 54 on which the electrostatic latent image is formed, whereby the electrostatic latent image is visualized to form a toner image.

The transfer roller 55 is arranged to face the photosensitive drum 54. A transfer bias is applied to the transfer roller 55 by bias application means (not shown). When the sheet S is nipped and conveyed between the transfer roller 55 and the photosensitive drum 54 on which the toner image is formed in a state where the transfer bias is applied to the surface of the transfer roller 55, the toner image formed on the surface of the photosensitive drum 54 is transferred to the surface of the sheet S. The sheet S on which the toner image has been transferred is conveyed to the fuser 6.

The fuser 6 has a heating unit 61 and a pressure roller 62, and fixes the image transferred to the sheet S by the process cartridge 50. The heating unit 61 is heated by being supplied with power from a power source (not shown). The pressure roller 62 is arranged to face the heating unit 61. One of the heating unit 61 and the pressure roller 62 is urged against the other by an urging mechanism (not shown), and the heating unit 61 and the pressure roller 62 are in close contact with each other.

When the sheet S on which the toner image has been transferred is conveyed to the fuser 6, the fuser 6 conveys and heats the sheet S while nipping the sheet S between the heating unit 61 and the pressure roller 62, thereby fixing the toner image on the sheet S. In this way, the fuser 6 conveys and heats the sheet S sent from the print engine 5.

The sheet discharging section 7 is located downstream of the print engine 5 in the sheet conveyance direction, and discharges the sheet S on which an image has been formed by the print engine 5 to the outside of the image forming apparatus 1. The sheet discharge section 7 includes a sheet discharge roller pair 71 and a sheet discharge tray 72. The sheet discharge roller pair 71 is configured to discharge the sheet S conveyed along the conveyance path P from the fuser 6 toward the outside of the apparatus main body 2. The sheet discharge tray 72 is formed on the upper surface of the apparatus main body 2, and supports the sheet S discharged to the outside of the apparatus main body 2 by the sheet discharge roller pair 71.

[Fuser]

As shown in FIGS. 2 and 3, the heating unit 61 of the fuser 6 includes a heater 611, a holder 612, a stay 613, and a belt

614. The heater **611** is a flat plate-shaped heater extending in the left-right direction. The heater **611** has a first surface **611A** and a second surface **611B** opposite to the first surface **611A**. The first surface **611A** is supported by the holder **612**.

The holder **612** is made of, for example, a resin member, and has a guide surface **612a** and a support wall **612b**. The guide surface **612** is in contact with an inner peripheral surface **614a** of the belt **614** and guides the belt **614**. The support wall **612b** has a support surface **612A** that supports the heater **611**. The support surface **612A** of the support wall **612b** is in contact with the first surface **611A** of the heater **611**. The stay **613** is a member that supports the holder **612**. The stay **613** is formed by bending a plate material having greater rigidity than the holder **612**, such as a steel plate, into a substantially U-shaped cross section.

The belt **614** is an endless belt having heat resistance and flexibility. The belt **614** has a metal tube made of metal such as stainless steel, and a fluororesin layer covering the metal tube. The heater **611**, the holder **612**, and the stay **613** are arranged inside the belt **614**. The belt **614** is configured to rotate around the heater **611**, the holder **612**, and the stay **613**. An inner peripheral surface **614a** of the belt **614** is in contact with the heater **611**.

The pressure roller **62** has a metal shaft **62A** and an elastic layer **62B** covering the shaft **62A**. The pressure roller **62** is pressed against the heater **611** via the belt **614**. The pressure roller **62** forms a nip portion NP for heating and pressing the sheet S by sandwiching the belt **614** with the heater **611**. That is, the pressure roller **62** heats and presses the sheet S together with the heater **611** at the nip portion NP.

The pressure roller **62** is configured to be rotationally driven by transmission of driving force from a motor **4** provided in the image forming apparatus **1**. Rotationally driving the pressure roller **62** causes the belt **614** to rotate followingly due to the frictional force between the pressure roller **62** and the belt **614** or the sheet S sandwiched at the nip portion NP. In this way, the sheet S on which a toner image has been transferred is conveyed between the pressure roller **62** and the heated belt **614**, and the toner image is thermally fixed. The pressure roller **62** is an example of a roller that conveys a sheet.

[Sheet Conveyance Device]

The sheet conveyance device T includes the motor **4**, a driving force transmission train **8**, the pressure roller **62**, and so on, and conveys the sheet S. The motor **4** is an example of a drive source.

[Driving Force Transmission Train]

As shown in FIG. **4**, the driving force transmission train **8** is a driving force transmission train for transmitting the driving force from the motor **4** to the pressure roller **62** of the fuser **6**. The driving force transmission train **8** is arranged at the left end of the apparatus main body **2**. The driving force transmission train **8** includes a motor gear **81**, a first idle gear **82**, a second idle gear **83**, a third idle gear **84**, a one-way clutch **90**, a fourth idle gear **85**, and a pressure roller gear **86**.

The motor gear **81** is fixed to an output shaft **4a** of the motor **4**, and the first idle gear **82** engages with the motor gear **81**. The second idle gear **83** engages with the first idle gear **82**, and the third idle gear **84** engages with the second idle gear **83**. The input side of the one-way clutch **90** is connected to the third idle gear **84**. The fourth idle gear **85** is connected to the output side of the one-way clutch **90**. The pressure roller gear **86** engages with the fourth idle gear **85**. The pressure roller gear **86** is fixed to the shaft **62A** of the pressure roller **62** so as to be integrally rotatable.

[One-Way Clutch]

As shown in FIGS. **5**, **6A** and **6B**, the one-way clutch **90** is configured to transmit the driving force from the motor **4** to the pressure roller **62**. The one-way clutch **90** includes a fixed shaft **91**, an input gear **92**, a transmission gear **93**, and an output gear **94**. The input gear **92** is the input-side gear of the one-way clutch **90**, and the output gear **94** is the output-side gear of the one-way clutch **90**.

In the one-way clutch **90**, the input gear **92**, the transmission gear **93**, and the output gear **94** are arranged in this order from left to right. That is, the transmission gear **93** is located between the input gear **92** and the output gear **94** in an axial direction X.

As shown in FIGS. **5** to **7**, the fixed shaft **91** is a long shaft member extending in the left-right direction, and has a first shaft portion **911** and a second shaft portion **912**. The direction along the left-right direction in which the fixed shaft **91** extends is the axial direction X of the fixed shaft **91**. The first shaft portion **911** has a first outer diameter D1, and the second shaft portion **912** has a second outer diameter D2 smaller than the first outer diameter D1 (D1>D2). The second shaft portion **912** is formed continuously to the right of the first shaft portion **911**.

As shown in FIGS. **5**, **6A**, **6B**, and **8**, the input gear **92** is rotatably supported by the first shaft portion **911** of the fixed shaft **91** and is rotatable by receiving the driving force from the motor **4**. The input gear **92** is rotatable in a first direction W1 and a second direction W2 opposite to the first direction W1. The input gear **92** has a body **920**, an input tooth portion **921**, a transmission portion **922** (transmission protrusion), an inclined cam surface **923**, and an inclined engagement surface **924**.

The body **920** is formed of a cylindrical member having a bottom portion **920a**, and the bottom portion **920a** is located at the left end of the body **920**. The input tooth portion **921** is formed in a cylindrical shape protruding leftward from the bottom portion **920a** of the body **920**, and input teeth **921a** are formed on the outer peripheral surface of the input tooth portion **921**. The input teeth **921a** engage with the third idle gear **84**. A driving force from the motor **4** is inputted to the input teeth **921a** via the motor gear **81**, the first idle gear **82**, the second idle gear **83**, and the third idle gear **84**.

The transmission portion **922** is a protrusion (protruding piece) that protrudes radially inward from the inner peripheral surface of the body **920**, and is configured to engage with the transmission gear **93**. In this embodiment, the transmission portions **922** are formed at two positions. The two transmission portions **922** are located at such positions that the distance from one transmission portion **922** to the other transmission portion **922** in a circumferential direction is equal to the distance from the other transmission portion **922** to the one transmission portion **922** in the circumferential direction.

The inclined cam surface **923** is an inclined surface formed on the bottom portion **920a** of the body **920** and facing the output gear **94** side in the axial direction X. Here, a phrase "the inclined cam surface **923** is an inclined surface facing the output gear **94** side" means that the inclined cam surface **923** is an inclined surface that is visible when viewed from the output gear **94** side in the axial direction X, that is, from the right side.

The inclined cam surface **923** extends along the circumferential direction. The inclined cam surface **923** is inclined relative to the direction perpendicular to the axial direction X such that the upstream side of the inclined cam surface **923** in the first direction W1 is closer to the output gear **94**

in the axial direction X than the downstream side. The inclined cam surface **923** has a downstream end **923a** in the first direction **W1** and an upstream end **923b** in the first direction **W1**. In the axial direction X, the upstream end **923b** is located closer to the output gear **94** than the downstream end **923a** is.

The inclined engagement surface **924** is formed inside the body **920**, and located upstream of the transmission portion **922** in the first direction **W1**. The inclined engagement surface **924** is an inclined surface facing away from the output gear **94** in the axial direction X. As shown in FIG. 9, a phrase “the inclined engagement surface **924** is an inclined surface facing away from the output gear **94**” means that the inclined engagement surface **924** is an inclined surface that is visible when viewed from the opposite side of the output gear **94** in the axial direction X, that is, from the left side (see the inclined engagement surface **924** surrounded by a dotted line in FIG. 9). Note that in FIG. 9, the bottom portion **920a** of the input gear **92** is shown to be transparent such that the inclined engagement surface **924** is visible.

The inclined engagement surface **924** extends along the circumferential direction. The inclined engagement surface **924** is inclined relative to the direction perpendicular to the axial direction X such that the downstream side of the inclined engagement surface **924** in the second direction **W2** is closer to the output gear **94** in the axial direction X than the upstream side. In this embodiment, the inclined engagement surfaces **924** are formed at two positions. The two inclined engagement surfaces **924** are arranged at such positions that the distance from one inclined engagement surface **924** to the other inclined engagement surface **924** in a circumferential direction is equal to the distance from the other inclined engagement surface **924** to the one inclined engagement surface **924** in the circumferential direction.

As shown in FIGS. 5, 6A, 6B, 10A and 10B, the transmission gear **93** is supported by the fixed shaft **91** so as to be movable in the axial direction X. The transmission gear **93** is rotatable in the first direction **W1** and the second direction **W2**. The transmission gear **93** has a cylindrical portion **930**, a transmitted portion **931** (transmitted protrusion), a contact surface **932**, a ratchet pawl **933**, and an engagement pawl **934**.

The cylindrical portion **930** is formed in a cylindrical shape. The cylindrical portion **930** has a small diameter portion **930a** at the end on the output gear **94** side in the axial direction X, the diameter of which is smaller than that of other portions. The transmitted portion **931** is a protrusion (protruding piece) that protrudes radially outward from the outer peripheral surface of the cylindrical portion **930**.

In this embodiment, the transmitted portions **931** are formed at two positions. The two transmitted portions **931** are arranged at such positions that the distance from one transmitted portion **931** to the other transmitted portion **931** in a circumferential direction is equal to the distance from the other transmitted portion **931** to the one transmitted portion **931** in the circumferential direction. The transmitted portion **931** is configured to engage with the transmission portion **922** of the input gear **92** when the input gear **92** rotates in the first direction **W1**.

The contact surface **932** is located at an end of the transmitted portion **931** facing the input gear **92** side in the axial direction X. The contact surface **932** is configured to contact the inclined cam surface **923** of the input gear **92** when the input gear **92** rotates in the first direction **W1**.

The ratchet pawl **933** protrudes toward the output gear **94** side from an end of the cylindrical portion **930** at the output gear **94** side in the axial direction X. The ratchet pawl **933**

has an engagement surface **933a** and a sliding surface **933b** formed continuously with the engagement surface **933a** in the circumferential direction. A plurality of ratchet pawls **933** are formed along the circumferential direction, and the engagement surfaces **933a** and the sliding surfaces **933b** are alternately arranged in the circumferential direction.

The engagement pawl **934** protrudes from the transmitted portion **931** toward the downstream side in the first direction **W1** in the circumferential direction. In this embodiment, the engagement pawls **934** are formed at two positions. The two engagement pawls **934** are arranged at such positions that the distance from one engagement pawl **934** to the other engagement pawl **934** in a circumferential direction is equal to the distance from the other engagement pawl **934** to the one engagement pawl **934** in the circumferential direction. The engagement pawl **934** is configured to engage with the inclined engagement surface **924** of the input gear **92** when the input gear **92** rotates in the second direction **W2**.

In a state where the transmission gear **93** is engaged with the input gear **92**, the transmission gear **93** is movable in the axial direction X between a first position where the small diameter portion **930a** of the cylindrical portion **930** is supported by the first shaft portion **911** of the fixed shaft **91** (the position shown in FIG. 6A) and a second position where the small diameter portion **930a** of the cylindrical portion **930** is supported by the second shaft portion **912** of the fixed shaft **91** (the position shown in FIG. 6B).

The small diameter portion **930a** of the transmission gear **93** located at the first position is supported by the first shaft portion **911** with a first sliding resistance between the small diameter portion **930a** and the first shaft portion **911**. The small diameter portion **930a** of the transmission gear **93** located at the second position is supported by the second shaft portion **912** with a second sliding resistance between the small diameter portion **930a** and the second shaft portion **912**. The second sliding resistance is smaller than the first sliding resistance.

In other words, the first shaft portion **911** of the fixed shaft **91** supports the transmission gear **93** located at the first position in a state where the first sliding resistance of a particular magnitude acts between the fixed shaft **91** and the transmission gear **93**. The second shaft portion **912** of the fixed shaft **91** supports the transmission gear **93** located at the second position in a state where the second sliding resistance smaller than the first sliding resistance acts between the fixed shaft **91** and the transmission gear **93**.

As shown in FIGS. 5, 6A, 6B, 11A and 11B, the output gear **94** is rotatably supported by the second shaft portion **912** of the fixed shaft **91**. The output gear **94** is connected to the pressure roller **62** of the fuser **6**, and transmits the driving force from the motor **4** to the pressure roller **62**. The output gear **94** is rotatable in the first direction **W1** and the second direction **W2**. The output gear **94** has a body **940**, ratchet recesses **941**, output teeth **942**, and spacers **943**.

The body **940** is formed in a cylindrical shape. The ratchet recess **941** is located at an end of the body **940** facing the input gear **92** in the axial direction X. The ratchet recess **941** is configured to engage with the ratchet pawl **933** of the transmission gear **93**. Specifically, the ratchet recess **941** engages with the ratchet pawl **933** when the transmission gear **93** rotates in the first direction **W1** relative to the output gear **94**, and does not engage with the ratchet pawl **933** when the transmission gear **93** rotates in the second direction **W2** relative to the output gear **94**.

The ratchet recess **941** has an engagement surface **941a** and a sliding surface **941b** formed continuously with the engagement surface **941a** in the circumferential direction. A

plurality of ratchet recesses **941** are formed along the circumferential direction, and the engagement surfaces **941a** and the sliding surfaces **941b** are alternately arranged in the circumferential direction.

The ratchet recess **941** is separated from the ratchet pawl **933** of the transmission gear **93** in the axial direction X when the transmission gear **93** is at the first position. The ratchet recess **941** is coupled to the ratchet pawl **933** of the transmission gear **93** in the axial direction X when the transmission gear **93** is at the second position.

When the transmission gear **93** rotates in the first direction W1 relative to the output gear **94** in a state where the transmission gear **93** is at the second position, the engagement surface **933a** of the ratchet pawl **933** and the engagement surface **941a** of the ratchet recess **941** are engaged and the transmission gear **93** and the output gear **94** rotate together.

When the transmission gear **93** rotates in the second direction W2 relative to the output gear **94** in a state where the transmission gear **93** is at the second position, the sliding surface **933b** of the ratchet pawl **933** slides relative to the sliding surface **941b** of the ratchet recess **941** in the circumferential direction, and the engagement surface **933a** and the engagement surface **941a** are disengaged. In this case, even if the transmission gear **93** rotates in the second direction W2, the output gear **94** does not rotate.

When the transmission gear **93** is at the first position, the ratchet recess **941** and the ratchet pawl **933** are separated from each other. Thus, the output gear **94** neither rotates when the transmission gear **93** rotates in the first direction W1 nor when the transmission gear **93** rotates in the second direction W2. When the output gear **94** rotates in a state where the transmission gear **93** is at the first position, the transmission gear **93** is not coupled to the output gear **94**.

The output teeth **942** are formed on the outer peripheral surface of the body **940** and engages with the fourth idle gear **85**. The output teeth **942** are coupled to the pressure roller **62** of the fuser **6** via the fourth idle gear **85** and pressure roller gear **86**. The output teeth **942** transmit the driving force from the motor **4** to the pressure roller **62** via the fourth idle gear **85** and pressure roller gear **86**.

The spacer **943** protrudes toward the input gear **92** from an end of the body **940** facing the input gear **92** side in the axial direction X, and is in contact with the input gear **92**. A plurality of spacers **943** are formed at intervals along the circumferential direction at the outer peripheral edge of the body **940**.

The spacer **943** regulates the position of the output gear **94** relative to the input gear **92** in the axial direction X by contacting the input gear **92**. By regulating the position of the output gear **94** relative to the input gear **92** with the spacer **943**, the transmission gear **93** is movable in the axial direction X between the input gear **92** and the output gear **94**. [Operation of One-Way Clutch]

Next, the operation of the one-way clutch **90** will be described.

In the one-way clutch **90** shown in FIG. 6A, the transmission gear **93** is located at the first position, and the contact surface **932** of the transmission gear **93** is in contact with the downstream end **923a** of the inclined cam surface **923** of the input gear **92**. The transmission gear **93** located at the first position is separated from the output gear **94** in the axial direction X.

In a state where the transmission gear **93** is located at the first position in this way, even if the output gear **94** rotates in the first direction W1 or the second direction W2, the output gear **94** and the transmission gear **93** are not coupled

to each other. Thus, in a state where the transmission gear **93** is located at the first position, even if the pressure roller **62** coupled to the output gear **94** rotates, the pressure roller **62** is not coupled to the motor **4** which is coupled to the input gear **92**.

When the transmission gear **93** is located at the first position, the small diameter portion **930a** of the cylindrical portion **930** of the transmission gear **93** is supported by the first shaft portion **911** of the fixed shaft **91** in a state where the first sliding resistance acts therebetween.

When the input gear **92** receives the driving force from the motor **4** and rotates in the first direction W1 in the state shown in FIG. 6A, the inclined cam surface **923** in contact with the contact surface **932** of the transmission gear **93** moves in the circumferential direction, and the transmission gear **93** receives force in the circumferential direction from the inclined cam surface **923** due to a sliding resistance between the transmission gear **93** and the inclined cam surface **923**.

The transmission gear **93** is supported by the first shaft portion **911** in a state where the first sliding resistance acts between the small diameter portion **930a** and the first shaft portion **911**. The first sliding resistance is set to be greater than the sliding resistance in the circumferential direction generated between the transmission gear **93** and the inclined cam surface **923**. Thus, when the input gear **92** rotates in the first direction W1 in a state where the contact surface **932** of the transmission gear **93** and the inclined cam surface **923** are in contact with each other, the transmission gear **93** keeps a state where rotation relative to the fixed shaft **91** is suppressed, and thus does not rotate at the same speed together with the input gear **92**.

As a result, the phase of the transmission gear **93** relative to the inclined cam surface **923** in the first direction W1 is shifted, and the position of the inclined cam surface **923** contacting the contact surface **932** of the transmission gear **93** is shifted from the downstream end **923a** to the upstream side in the first direction W1. The upstream side of the inclined cam surface **923** in the first direction W1 is located closer to the output gear **94** than the downstream side of the inclined cam surface **923** is. Thus, when the position of the inclined cam surface **923** contacting the contact surface **932** of the transmission gear **93** moves to the upstream side in the first direction W1, the transmission gear **93** is pressed by the inclined cam surface **923** toward the output gear **94** in the axial direction X and moves.

As shown in FIG. 6B, when the position of the inclined cam surface **923** contacting the contact surface **932** of the transmission gear **93** reaches the upstream end **923b**, the transmission gear **93** moves to the second position.

In this way, when the input gear **92** rotates in the first direction W1, the first sliding resistance between the small diameter portion **930a** and the first shaft portion **911** prevents the transmission gear **93** located at the first position from rotating at the same speed together with the input gear **92**, and thus enables the transmission gear **93** to move from the first position to the second position.

After the transmission gear **93** has moved to the second position, the transmission portion **922** of the input gear **92** rotating in the first direction W1 contacts the transmitted portion **931** of the transmission gear **93**. When the transmission portion **922** contacts the transmitted portion **931**, a force in the first direction W1 acts on the transmission gear **93** from the input gear **92**, and the input gear **92** and the transmission gear **93** rotate together in the first direction W1.

The second sliding resistance between the second shaft portion **912** and the small diameter portion **930a** of the

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transmission gear 93 located at the second position is smaller than the first sliding resistance. Thus, the transmission gear 93 rotates smoothly against the second sliding resistance.

In the one-way clutch 90, the first outer diameter D1 of the first shaft portion 911 for supporting the transmission gear 93 located at the first position is different from the second outer diameter D2 of the second shaft portion 912 for supporting the transmission gear 93 located at the second position. This causes the magnitude of the first sliding resistance and the magnitude of the second sliding resistance be different from each other, thereby simplifying the configuration of the one-way clutch 90.

When the transmission gear 93 is located at the second position and rotates in the first direction W1, the engagement surface 933a of the ratchet pawl 933 of the transmission gear 93 engages with the engagement surface 941a of the ratchet recess 941 of the output gear 94, and the transmission gear 93 and the output gear 94 rotate integrally in the first direction W1.

In this way, the input gear 92 receives the driving force from the motor 4 and rotates in the first direction W1, thereby moving the transmission gear 93 from the first position to the second position, so that the transmission gear 93 is coupled to the output gear 94 and the driving force in the first direction W1 from the motor 4 is transmitted to the output gear 94 via the transmission gear 93.

In other words, when the input gear 92 rotates in the first direction W1, the transmission gear 93 located at the second position and coupled to the output gear 94 rotates relative to the fixed shaft 91 together with the input gear 92 against the second sliding resistance. This enables the driving force to be transmitted from the transmission gear 93 to the output gear 94.

In the one-way clutch 90, when the input gear 92 rotates in the first direction W1, the transmission gear 93 is pressed toward the output gear 94 by the inclined cam surface 923 and moves from the first position to the second position. This enables switching from a state where the transmission of the driving force to the output gear 94 is canceled to a state where the driving force is transmitted to the output gear 94 via the transmission gear 93 with a simple configuration.

Conversely, in the one-way clutch 90, when the input gear 92 receives the driving force from the motor 4 and rotates in the second direction W2 from the state where the transmission gear 93 is located at the second position shown in FIG. 6B, the transmission gear 93 is moved from the second position to the first position so as to cancel transmission of the driving force to the output gear 94.

Specifically, when the input gear 92 receives the driving force from the motor 4 and rotates in the second direction W2 in a state where the transmission gear 93 is at the second position, the inclined engagement surface 924 of the input gear 92 engages with the engagement pawl 934 of the transmission gear 93. The downstream side of the inclined engagement surface 924 in the second direction W2 is located closer to the output gear 94 than the upstream side of the inclined engagement surface 924 in the second direction W2. Thus, when the engagement pawl 934 engages with the inclined engagement surface 924 and is pressed in the second direction W2, the position of the inclined engagement surface 924 contacting the engagement pawl 934 moves upstream in the second direction W2, and the transmission gear 93 is pulled toward the input gear 92 in the axial direction X by the inclined engagement surface 924.

In other words, when the transmission gear 93 is located at the second position and the input gear 92 rotates in the

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second direction W2, the transmission gear 93 is pulled toward the input gear 92 in the axial direction X by the inclined engagement surface 924 and moves to the first position. As a result, the transmission gear 93 and the output gear 94 are separated from each other, and transmission of the driving force from the input gear 92 to the output gear 94 is cancelled.

As described above, in the one-way clutch 90, by rotating the input gear 92 in the second direction, the inclined engagement surface 924 and the engagement pawl 934 are engaged, and the transmission gear 93 is moved from the second position to the first position.

In the one-way clutch 90, when the transmission gear 93 is at the first position, the transmission gear 93 and the output gear 94 are separated without being coupled. Thus, even when the output gear 94 is rotated, the transmission gear 93 does not rotate together with the output gear 94 and the output gear 94 idles.

Thus, the load of the motor 4 and the gear train between the motor 4 and the input gear 92 is not applied to the output gear 94 coupled to the pressure roller 62. This enables the pressure roller 62 to be easily rotated when removing a jammed sheet S, without providing a complicated mechanism such as a driving force transmission canceling mechanism in addition to the one-way clutch 90 in the driving force transmission train 8. Thus, removing the jammed sheet S is performed smoothly.

When the transmission gear 93 is at the first position, the one-way clutch 90 cuts off the transmission of the driving force from the motor 4 to the pressure roller 62. Thus, even if the pressure roller 62 is rotated when removing a jammed sheet, the load of the motor 4 is not applied to the pressure roller 62, and a process of removing a jammed sheet is performed smoothly.

Since the one-way clutch 90 is coupled to the pressure roller 62 of the fuser 6, the sheet S is smoothly removed from the fuser 6 when a jam of the sheet S occurs in the fuser 6.

The heating unit 61 of the fuser 6 has the heater 611 of a flat plate shape, and the pressure roller 62 is pressed against the heater 611. Thus, the rotational torque of the pressure roller 62 tends to be larger than a case where the heating unit 61 is formed by a roller member. However, in the one-way clutch 90, by moving the transmission gear 93 to the first position where the transmission gear 93 and the output gear 94 are separated from each other, the load applied to the pressure roller 62 does not increase and a jammed sheet is removed smoothly.

[Motor Drive Control When a Jam Occurs]

As shown in FIG. 12, the image forming apparatus 1 includes a controller C and the sheet discharge sensor 98. The controller C is connected to the motor 4 and configured to control the operation of the motor 4. As shown in FIG. 1, the sheet discharge sensor 98 is arranged between the fuser 6 and the sheet discharge roller pair 71 in the sheet conveyance direction, and is configured to detect the sheet S that passes through the fuser 6.

The controller C is configured to, when the detection of the sheet S by the sheet discharge sensor 98 continues for a particular time or longer, for example, determine that the sheet S is jammed in the fuser 6. In response to determining that the sheet S is jammed, the controller C controls the motor 4 to rotate in the direction in which the input gear 92 rotates in the second direction W2.

When the motor 4 is driven to rotate the input gear 92 in the second direction W2, the inclined engagement surface 924 engages with the engagement pawl 934, the transmis-

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sion gear 93 moves to the first position, and the transmission gear 93 separates from the output gear 94.

In this way, when the sheet S is jammed, the motor 4 is driven under the control of the controller C to rotate the input gear 92 in the second direction W2, thereby moving the transmission gear 93 from the second position to the first position so that the transmission gear 93 and the output gear 94 are separated from each other. Thus, when the pressure roller 62 is rotated for removing a jam, the load of the motor 4 is not applied to the pressure roller 62, and the jam is removed smoothly.

While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Thus, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described invention are provided below:

For example, in response to determining that a sheet S is jammed at a position other than the fuser 6, the controller C may control the motor 4 to rotate in a direction in which the input gear 92 rotates in the second direction W2.

What is claimed is:

1. A sheet conveyance device comprising:

a motor;

a roller configured to convey a sheet; and

a one-way clutch configured to transmit a driving force from the motor to the roller, the one-way clutch including:

a fixed shaft extending in an axial direction;

an input gear rotatably supported by the fixed shaft, the input gear being rotatable by receiving the driving force from the motor;

an output gear rotatably supported by the fixed shaft, the output gear being coupled to the roller and configured to transmit the driving force to the roller; and

a transmission gear supported by the fixed shaft, the transmission gear being movable in the axial direction between a first position and a second position in a state where the transmission gear engages with the input gear, the first position being a position at which the transmission gear is separated from the output gear, the second position being a position at which the transmission gear is coupled to the output gear,

the input gear being configured to:

by rotating in a first direction by receiving the driving force from the motor, move the transmission gear from the first position to the second position and transmit the driving force in the first direction to the output gear via the transmission gear; and

by rotating in a second direction opposite the first direction by receiving the driving force from the motor, move the transmission gear from the second position to the first position and cancel transmission of the driving force to the output gear,

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the transmission gear being configured to, when the output gear rotates in a state where the transmission gear is located at the first position, not be coupled to the output gear,

wherein the fixed shaft is configured to:

support the transmission gear located at the first position in a state where a first sliding resistance of a particular magnitude acts between the fixed shaft and the transmission gear; and

support the transmission gear located at the second position in a state where a second sliding resistance acts between the fixed shaft and the transmission gear, the second sliding resistance being smaller than the first sliding resistance.

2. The sheet conveyance device according to claim 1, wherein, when the roller rotates in a state where the transmission gear is located at the first position, the roller is not coupled to the motor.

3. The sheet conveyance device according to claim 1,

wherein the transmission gear located at the first position is configured to:

when the input gear rotates in the first direction, be moved by the input gear from the first position to the second position by keeping a state where rotation relative to the fixed shaft is suppressed due to the first sliding resistance; and

wherein the transmission gear located at the second position is configured to:

when the input gear rotates in the first direction, rotate relative to the fixed shaft together with the input gear against the second sliding resistance.

4. The sheet conveyance device according to claim 1, wherein the fixed shaft includes:

a first shaft portion having a first outer diameter; and

a second shaft portion having a second outer diameter smaller than the first outer diameter;

wherein the first shaft portion supports the transmission gear located at the first position in a state where the first sliding resistance acts between the first shaft portion and the transmission gear; and

wherein the second shaft portion supports the transmission gear located at the second position in a state where the second sliding resistance acts between the second shaft portion and the transmission gear.

5. The sheet conveyance device according to claim 1, wherein the transmission gear is located between the input gear and the output gear in the axial direction;

wherein the input gear includes:

input teeth to which the driving force is inputted;

a transmission protrusion engageable with the transmission gear; and

an inclined cam surface facing toward the output gear in the axial direction, the inclined cam surface extending along a circumferential direction, an upstream side of the inclined cam surface in the first direction being closer to the output gear in the axial direction than a downstream side of the inclined cam surface in the first direction is;

wherein the transmission gear includes:

a transmitted protrusion engageable with the transmission protrusion when the input gear rotates in the first direction;

a contact surface configured to contact the inclined cam surface when the input gear rotates in the first direction; and

a ratchet pawl protruding toward the output gear in the axial direction; and

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wherein the output gear includes:

a ratchet recess configured to:

- engage with the ratchet pawl when the transmission gear rotates in the first direction; and
- not engage with the ratchet pawl when the transmission gear rotates in the second direction; and

output teeth coupled to the roller.

6. The sheet conveyance device according to claim 5, wherein the input gear includes:

- an inclined engagement surface facing away from the output gear in the axial direction, the inclined engagement surface extending in a circumferential direction, a downstream side of the inclined engagement surface in the second direction being closer to the output gear in the axial direction than an upstream side of the inclined engagement surface in the second direction is; and

wherein the transmission gear includes:

- an engagement pawl configured to engage with the inclined engagement surface when the input gear rotates in the second direction.

7. The sheet conveyance device according to claim 1, further comprising a controller configured to, in response to determining that a jam of a sheet occurs, control the motor to drive in a direction in which the input gear rotates in the second direction.

8. The sheet conveyance device according to claim 1, wherein the output gear includes a body and a spacer, the spacer protruding from the body toward the input gear in the axial direction, the spacer being in contact with the input gear to regulate a position of the output gear relative to the input gear in the axial direction; and

wherein the transmission gear is located between the input gear and the output gear in the axial direction, the transmission gear being movable in the axial direction between the input gear and the output gear.

9. An image forming apparatus comprising:

a print engine configured to form an image on a sheet; a fuser configured to convey and heat the sheet conveyed from the print engine, the fuser including a roller; a motor; and

a one-way clutch configured to transmit a driving force from the motor to the roller, the one-way clutch including:

a fixed shaft extending in an axial direction;

an input gear rotatably supported by the fixed shaft, the input gear being rotatable by receiving the driving force from the motor;

an output gear rotatably supported by the fixed shaft, the output gear being coupled to the roller and configured to transmit the driving force to the roller; and

a transmission gear supported by the fixed shaft, the transmission gear being movable in the axial direction between a first position and a second position in a state where the transmission gear engages with the input gear, the first position being a position at which the transmission gear is separated from the output gear, the second position being a position at which the transmission gear is coupled to the output gear, the input gear being configured to:

by rotating in a first direction by receiving the driving force from the motor, move the transmission gear from the first position to the second position and transmit the driving force in the first direction to the output gear via the transmission gear; and

by rotating in a second direction opposite the first direction by receiving the driving force from the

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motor, move the transmission gear from the second position to the first position and cancel transmission of the driving force to the output gear,

the transmission gear being configured to, when the output gear rotates in a state where the transmission gear is located at the first position, not be coupled to the output gear,

wherein the fixed shaft is configured to:

support the transmission gear located at the first position in a state where a first sliding resistance of a particular magnitude acts between the fixed shaft and the transmission gear; and

support the transmission gear located at the second position in a state where a second sliding resistance acts between the fixed shaft and the transmission gear, the second sliding resistance being smaller than the first sliding resistance.

10. The image forming apparatus according to claim 9, wherein the fuser includes a heater having a flat plate shape; and

wherein the roller is a pressure roller configured to heat and press the sheet in cooperation with the heater.

11. The image forming apparatus according to claim 9, wherein, when the roller rotates in a state where the transmission gear is located at the first position, the roller is not coupled to the motor.

12. The image forming apparatus according to claim 9, wherein the transmission gear located at the first position is configured to:

when the input gear rotates in the first direction, be moved by the input gear from the first position to the second position by keeping a state where rotation relative to the fixed shaft is suppressed due to the first sliding resistance; and

wherein the transmission gear located at the second position is configured to:

when the input gear rotates in the first direction, rotate relative to the fixed shaft together with the input gear against the second sliding resistance.

13. The image forming apparatus according to claim 9, wherein the fixed shaft includes:

a first shaft portion having a first outer diameter; and

a second shaft portion having a second outer diameter smaller than the first outer diameter;

wherein the first shaft portion supports the transmission gear located at the first position in a state where the first sliding resistance acts between the first shaft portion and the transmission gear; and

wherein the second shaft portion supports the transmission gear located at the second position in a state where the second sliding resistance acts between the second shaft portion and the transmission gear.

14. The image forming apparatus according to claim 9, wherein the transmission gear is located between the input gear and the output gear in the axial direction;

wherein the input gear includes:

input teeth to which the driving force is inputted;

a transmission protrusion engageable with the transmission gear; and

an inclined cam surface facing toward the output gear in the axial direction, the inclined cam surface extending along a circumferential direction, an upstream side of the inclined cam surface in the first direction being closer to the output gear in the axial direction than a downstream side of the inclined cam surface in the first direction is;

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wherein the transmission gear includes:

a transmitted protrusion engageable with the transmission protrusion when the input gear rotates in the first direction;

a contact surface configured to contact the inclined cam surface when the input gear rotates in the first direction; and

a ratchet pawl protruding toward the output gear in the axial direction; and

wherein the output gear includes:

a ratchet recess configured to:

engage with the ratchet pawl when the transmission gear rotates in the first direction; and

not engage with the ratchet pawl when the transmission gear rotates in the second direction; and

output teeth coupled to the roller.

15. The image forming apparatus according to claim 14, wherein the input gear includes:

an inclined engagement surface facing away from the output gear in the axial direction, the inclined engagement surface extending in a circumferential direction, a downstream side of the inclined engagement surface in the second direction being closer to the output gear in the axial direction than an upstream side of the inclined engagement surface in the second direction is; and

wherein the transmission gear includes:

an engagement pawl configured to engage with the inclined engagement surface when the input gear rotates in the second direction.

16. The image forming apparatus according to claim 9, further comprising a controller configured to, in response to determining that a jam of a sheet occurs, control the motor to drive in a direction in which the input gear rotates in the second direction.

17. The image forming apparatus according to claim 9, wherein the output gear includes a body and a spacer, the spacer protruding from the body toward the input gear in the axial direction, the spacer being in contact with the input gear to regulate a position of the output gear relative to the input gear in the axial direction; and

wherein the transmission gear is located between the input gear and the output gear in the axial direction, the transmission gear being movable in the axial direction between the input gear and the output gear.

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18. A sheet conveyance device comprising:

a motor;

a roller configured to convey a sheet; and

a one-way clutch configured to transmit a driving force from the motor to the roller, the one-way clutch including:

a fixed shaft extending in an axial direction;

an input gear rotatably supported by the fixed shaft, the input gear being rotatable by receiving the driving force from the motor;

an output gear rotatably supported by the fixed shaft, the output gear being coupled to the roller and configured to transmit the driving force to the roller; and

a transmission gear supported by the fixed shaft, the transmission gear being movable in the axial direction between a first position and a second position in a state where the transmission gear engages with the input gear, the first position being a position at which the transmission gear is separated from the output gear, the second position being a position at which the transmission gear is coupled to the output gear,

the input gear being configured to:

by rotating in a first direction by receiving the driving force from the motor, move the transmission gear from the first position to the second position and transmit the driving force in the first direction to the output gear via the transmission gear; and

by rotating in a second direction opposite the first direction by receiving the driving force from the motor, move the transmission gear from the second position to the first position and cancel transmission of the driving force to the output gear,

the transmission gear being configured to, when the output gear rotates in a state where the transmission gear is located at the first position, not be coupled to the output gear,

wherein the fixed shaft includes:

a first shaft portion having a first outer diameter; and a second shaft portion having a second outer diameter smaller than the first outer diameter;

wherein the first shaft portion supports the transmission gear located at the first position; and

wherein the second shaft portion supports the transmission gear located at the second position.

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